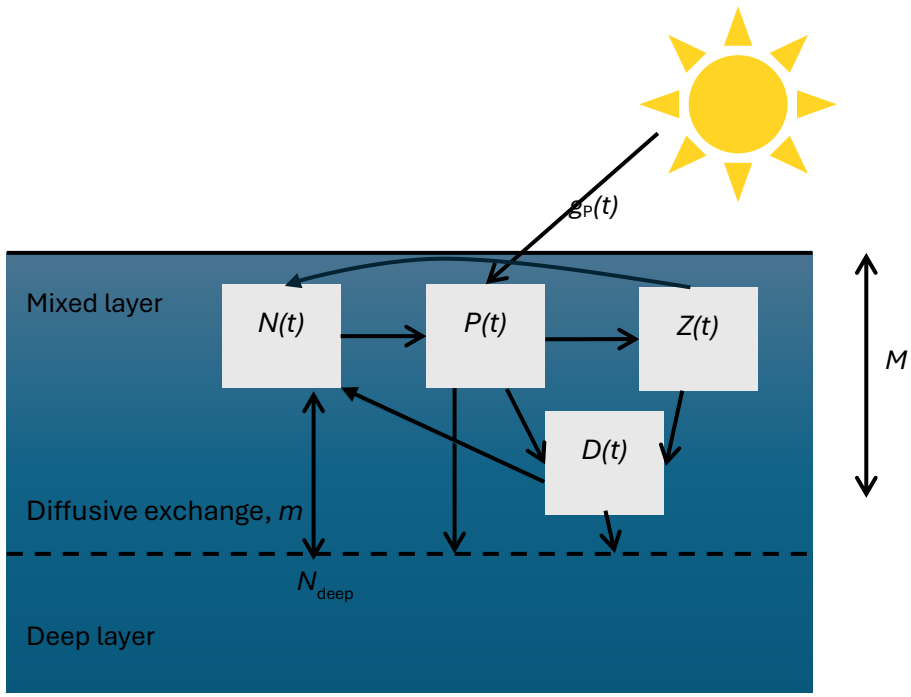
Exercise 1

This exercise is about setting up a model of the Nutrient-Phytoplankton-Zooplankton-Detritus (NPZD) dynamics in coastal waters. The water column is represented as a chemostat and only the state variables in the upper mixed layer are resolved.



The fundamental processes are the growth of phytoplankton due to light and nutrients, and the predation of zooplankton on phytoplankton. Each of these processes are describes as rates (units of 1/day):

Phytoplankton growth (1/day):

$$g_{\mathrm{P}} = g_{\mathrm{P.max}} \min\left\{\frac{L}{L + k_{PAR}}, \frac{N}{N + k_N}\right\}$$

Zooplankton consumption (1/day):

$$g_{\mathrm{Z}} = g_{\mathrm{Z.max}} \frac{P}{P + k_{\mathrm{Z}}}$$

State variables in the upper mixed layer. The red terms are those associated with mixing into the deep layer or to sinking.

$$\frac{\mathrm{d}N}{\mathrm{d}t} = -g_{\mathrm{P}}P + \epsilon_{\mathrm{N}} g_{\mathrm{Z}} Z + rD + \left(N_{\mathrm{deep}} - N\right)\frac{\kappa}{M}$$

$$\frac{\mathrm{d}P}{\mathrm{d}t} = g_{\mathrm{P}}P - g_{\mathrm{Z}}Z - m_{\mathrm{P}}P - \frac{\kappa}{M}P$$

$$\frac{\mathrm{d}Z}{\mathrm{d}t} = (1 - \epsilon_{\mathrm{N}} - \epsilon_{\mathrm{D}}) g_{\mathrm{Z}} Z - m_{\mathrm{Z}} Z$$

$$\frac{\mathrm{d}D}{\mathrm{d}t} = m_{\mathrm{P}}P + m_{\mathrm{Z}}Z + \epsilon_{\mathrm{D}} g_{\mathrm{Z}} Z - rD - \frac{\kappa}{M}D - \frac{w_{\mathrm{D}}}{M}D$$

Parameters:

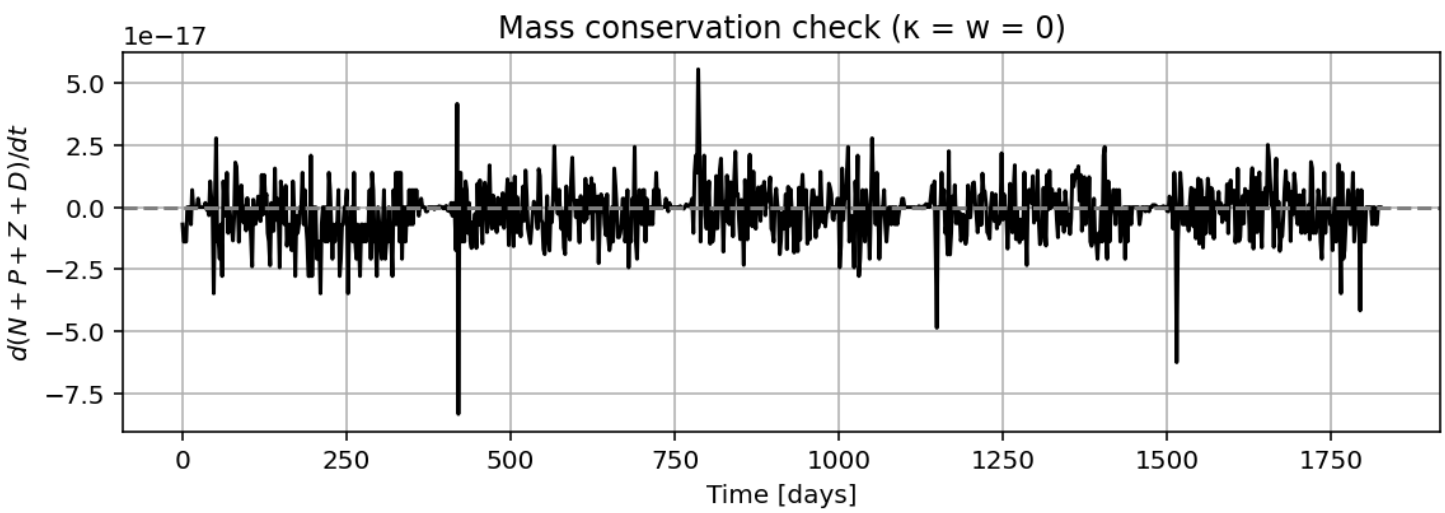
Name	Symbol and value	Units
Deep nutrient concentration	$N_{\mathrm{deep}} = 10$	mmol N m^-3
Maximum phytoplankton growth rate	$g_{\mathrm{Pmax}} = 2.0$	d^-1
Light half-saturation constant	$k_{\mathrm{L}} = 20$	W m^-2
Nutrient half-saturation constant	$k_{\mathrm{N}} = 0.3$	mmol N m^-3
Phytoplankton mortality rate	$m_{\mathrm{P}} = 0.07$	d^-1
Maximum grazing rate	$g_{\mathrm{Zmax}} = 1.0$	d^-1
Grazing half-saturation constant	$k_{\mathrm{Z}} = 1.0$	mmol N m^-3
Respiration fraction (eN)	$\epsilon_{\mathrm{resp}} = 0.3$	-
Egestion efficiency (eD)	$\epsilon_{\mathrm{egestion}} = 0.3$	-
Zooplankton mortality rate	$m_{\mathrm{Z}} = 0.2$	d^-1
Remineralization rate (r)	$\tau = 0.1$	d^-1
Detritus sinking speed	$w_{\mathrm{D}} = 2$	m d^-1
Vertical exchange rate	$\kappa = 0.3$	m d^-1
Mixed-layer depth	$M = 60$	m

Every RHS term has units: mmol N m^-3 d^-1

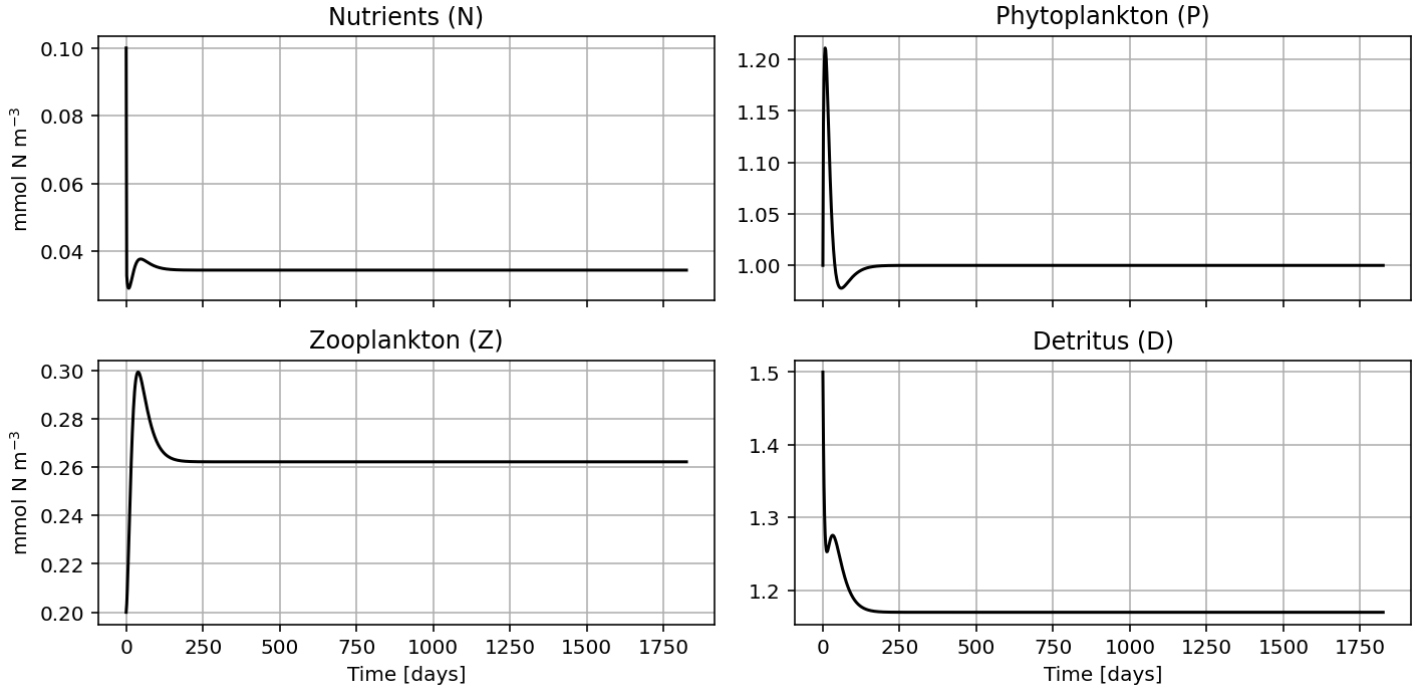
Exercise:

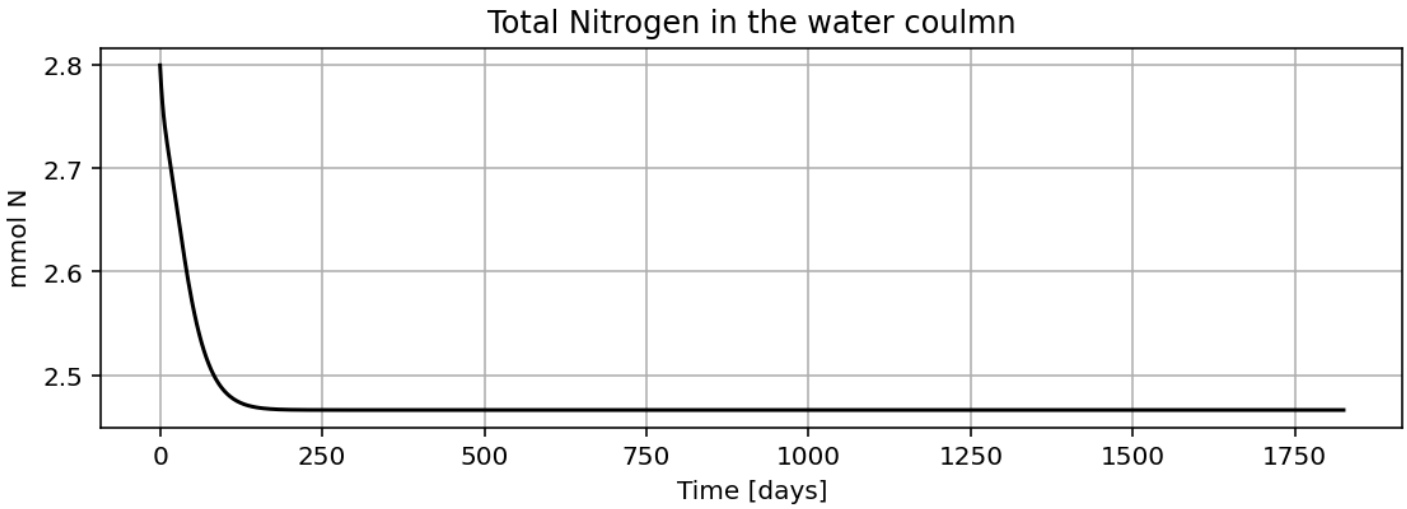
1. Identify the role of each term in the equations
2. Identify the name and units of each term in the table with parameters
3. Code a function that given time and a vector with the 4 state variables returns the sum of all the derivatives. Use this function to confirm that the model conserves mass in the absence of mixing and sinking ($m = w_{\mathrm{D}} = 0$).
4. Use a stiff ODE solver to integrate the model until convergence with constant light. Plot the four state variables as a function of time and the total amount of nitrogen in the water column.
5. Integrate the model with time-varying between 0 and 20 mmol photons/m2/day. How many years does it take to converge.

3.)

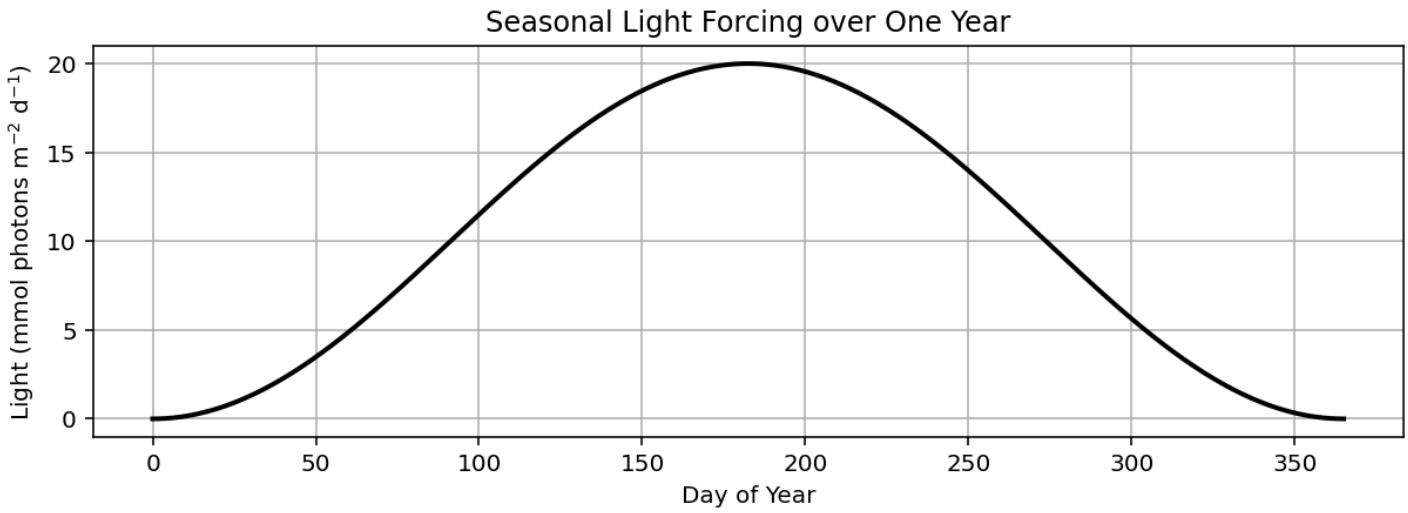


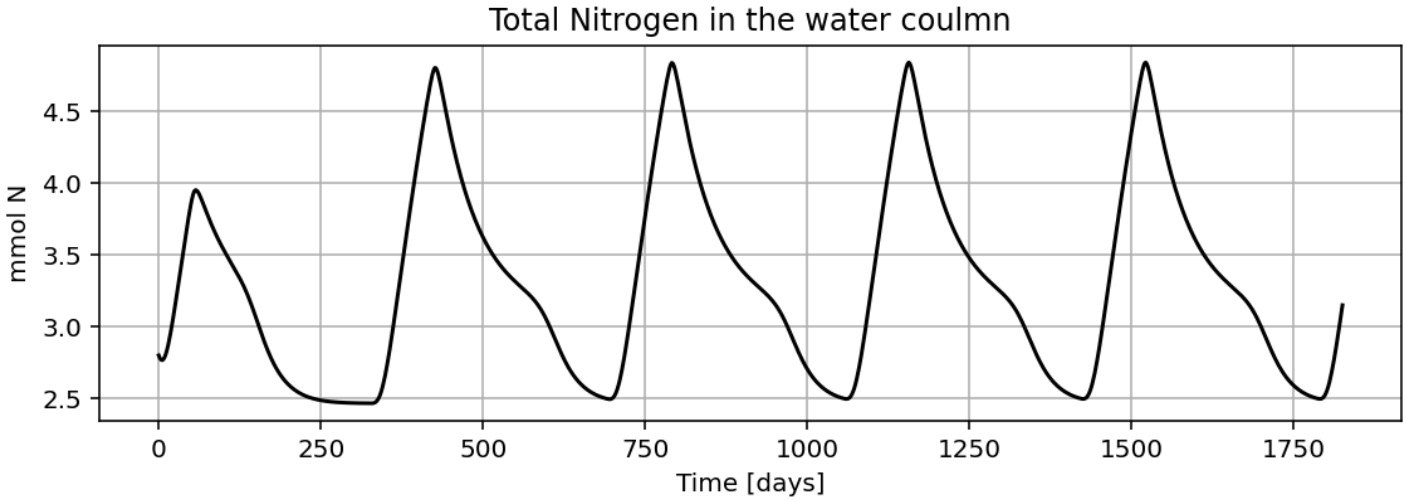
4.)


Nutrients (N)
Phytoplankton (P)
Zooplankton (Z)
Detritus (D)
mmol N m^{-3}
Time [days]

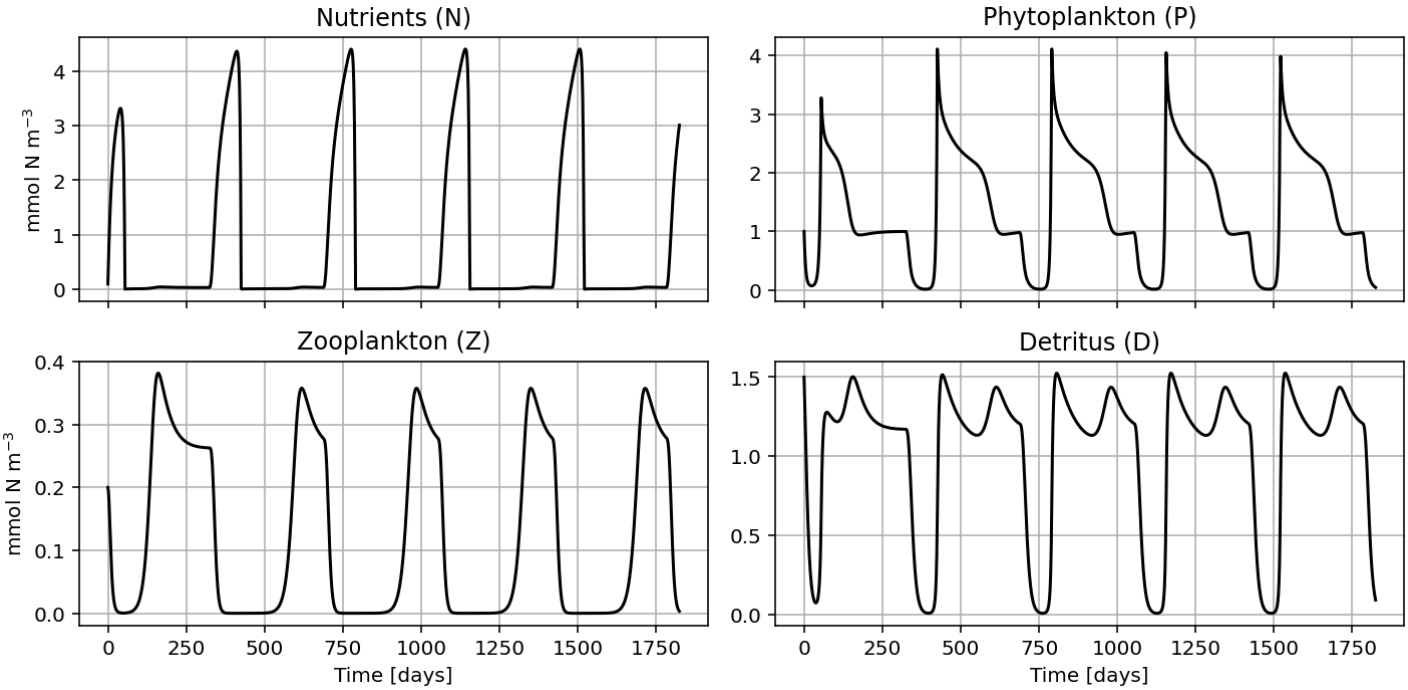

Total Nitrogen in the water coulmn
mmol N
Time [days]

5.)


Seasonal Light Forcing over One Year
Light (mmol photons m^{-2} d^{-1})
Day of Year



It very much converges after the second year



Python-Code

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.integrate import solve_ivp


time_varying = True
# ————————————————————
# PARAMETERS (from your model)
# ————————————————————
N_deep = 10.0        # mmol N m^-3
P_growth_max = 2.0   # d^-1
Z_consum_max = 1.0   # d^-1

k_L = 20.0           # light half-sat
k_N = 0.3            # mmol N m^-3
k_Z = 1.0            # mmol N m^-3

m_P = 0.07           # d^-1
m_Z = 0.2            # d^-1

e_N = 0.3            # -
e_D = 0.3            # -

r = 0.1              # d^-1
w_D = 2.0            # m d^-1

kappa = 0.3          # m d^-1
M = 60.0             # m

If time_varying:
    L_max = 20.0  # mmol photons/m^2/day
    def light(t):
        # Seasonal sinusoidal forcing: varies 0 -> L_max
        return L_max * (0.5 + 0.5 * np.sin(2 * np.pi * t / 365 - np.pi/2))
else:
    L = 50

# ————————————————————
# TIME SETUP
# ————————————————————
t_max = 365 * 5
t_span = (0, t_max)          # days
t_eval = np.linspace(0, t_max, 1000)

# ————————————————————
# INITIAL CONDITIONS
# ————————————————————
N0 = 0.1
P0 = 1.0
Z0 = 0.2
D0 = 1.5

y0 = [N0, P0, Z0, D0]

# ————————————————————
# ODE SYSTEM
# ————————————————————
def rhs(t, y):
    N, P, Z, D = y

    L_current = light(t) If time_varying else L

    P_growth = P_growth_max * min((L_current / (L_current + k_L)),
                                  (N / (N + k_N)))
    Z_consum = Z_consum_max * (P / (P + k_Z))

    N_uptake  = P_growth * P
    P_mort    = m_P * P
    Z_grazing = Z_consum * Z
    Z_mort    = m_Z * Z
    remin     = r * D

    dN = -N_uptake + e_N * Z_grazing + remin + (N_deep - N) * (kappa / M)
    dP =  N_uptake - Z_grazing - P_mort - (kappa / M) * P
    dZ = (1 - e_N - e_D) * Z_grazing - Z_mort
    dD = P_mort + e_D * Z_grazing + Z_mort - remin - ((kappa / M) + (w_D / M)) * D

    return [dN, dP, dZ, dD]

# ————————————————————
# SOLVE ODEs
# ————————————————————
sol = solve_ivp(
    rhs,
    t_span,
    y0,
    t_eval=t_eval,
    method="RK45",
    rtol=1e-6,
    atol=1e-9
)

# ————————————————————————————
# TOTAL MASS DERIVATIVE FUNCTION
# ————————————————————————————
def total_mass_derivative(t, y):
    """
    Returns d/dt (N + P + Z + D)
    """
    return np.sum(rhs(t, y))
# ————————————————————————————
# EVALUATE MASS CONSERVATION
# ————————————————————————————
dMdt = np.array([
    total_mass_derivative(t, sol.y[:, i])
    for i, t in enumerate(sol.t)
])
total_mass = sol.y.sum(axis=0)

# ————————————————————————————
# PLOTS
# ————————————————————————————
# Total mass tendency
plt.figure(figsize=(8, 3))
plt.plot(sol.t, dMdt, color="black")
plt.axhline(0, color="gray", linestyle="--")
plt.xlabel("Time [days]")
plt.ylabel(r"$d(N+P+Z+D)/dt$")
plt.title("Mass conservation check (κ = w = 0)")
plt.grid(True)
plt.tight_layout()
plt.show()

# Total mass
plt.figure(figsize=(8, 3))
plt.plot(sol.t, total_mass, color="black")
plt.xlabel("Time [days]")
plt.ylabel("mmol N")
plt.title("Total Nitrogen in the water coulmn")
plt.grid(True)
plt.tight_layout()
plt.show()

# ————————————————————
# SIMPLE PLOT
# ————————————————————
fig, axs = plt.subplots(2, 2, figsize=(10, 5), sharex=True)

axs[0, 0].plot(sol.t, sol.y[0], color="black")
axs[0, 0].set_title("Nutrients (N)")
axs[0, 0].set_ylabel("mmol N m$^{-3}$")

axs[0, 1].plot(sol.t, sol.y[1], color="black")
axs[0, 1].set_title("Phytoplankton (P)")

axs[1, 0].plot(sol.t, sol.y[2], color="black")
axs[1, 0].set_title("Zooplankton (Z)")
axs[1, 0].set_xlabel("Time [days]")
axs[1, 0].set_ylabel("mmol N m$^{-3}$")

axs[1, 1].plot(sol.t, sol.y[3], color="black")
axs[1, 1].set_title("Detritus (D)")
axs[1, 1].set_xlabel("Time [days]")

for ax in axs.flat:
    ax.grid(True)

plt.tight_layout()
plt.show()
```